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JEE Main Physics Practice Sheet — DPP-048

Topic: Centre of Mass of a Rigid Body & Continuous Mass Distribution**Time Allowed:** 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A thin rod of length L lies along the x -axis with one end at the origin. Its linear mass density varies as $\lambda(x) = \lambda_0 \left(\frac{x}{L}\right)$. The x -coordinate of its centre of mass is:

- (A) $\frac{2}{3}L$
 (B) $\frac{1}{2}L$
 (C) $\frac{3}{4}L$
 (D) $\frac{1}{3}L$

Q2. The distance of the centre of mass of a uniform semi-circular thin wire of radius R from its geometric centre along the axis of symmetry is:

- (A) $\frac{4R}{3\pi}$
 (B) $\frac{R}{\pi}$
 (C) $\frac{2R}{\pi}$
 (D) $\frac{3R}{8}$

Q3. From a uniform circular disc of radius R , a circular hole of radius $R/2$ is cut out such that the hole touches the outer edge of the original disc. The distance of the centre of mass of the remaining portion from the centre of the original disc is:

- (A) $\frac{R}{3}$
 (B) $\frac{R}{4}$
 (C) $\frac{R}{8}$
 (D) $\frac{R}{6}$

Q4. The centre of mass of a uniform solid hemisphere of radius R lies on its central axis of symmetry at a distance y_{cm} from the flat base. The value of y_{cm} is:

- (A) $\frac{3R}{8}$
 (B) $\frac{R}{2}$
 (C) $\frac{4R}{3\pi}$
 (D) $\frac{2R}{3\pi}$

Q5. A uniform wire of total length $3L$ is bent to form an equilateral triangle. If one vertex is at the origin and one side lies along the x -axis from $x = 0$ to $x = L$, the y -coordinate of the centre of mass is:

- (A) $\frac{L}{2\sqrt{3}}$
 (B) $\frac{\sqrt{3}}{2}L$
 (C) $\frac{L}{3\sqrt{3}}$
 (D) $\frac{\sqrt{3}}{6}L$

Q6. The centre of mass of a uniform hollow hemispherical shell of radius R lies at a distance from its circular base equal to:

- (A) $\frac{R}{2}$
 (B) $\frac{3R}{8}$
 (C) $\frac{2R}{\pi}$
 (D) $\frac{4R}{3\pi}$

Q7. A thin rod of length L has non-uniform linear mass density $\lambda(x) = \lambda_0 \left(1 + \frac{x}{L}\right)$ for $0 \leq x \leq L$. The x -coordinate of its centre of mass is:

- (A) $\frac{7}{12}L$
 (B) $\frac{5}{9}L$
 (C) $\frac{9}{16}L$
 (D) $\frac{3}{5}L$

Q8. The position of the centre of mass of a uniform solid right circular cone of height h and base radius R from its apex along the central axis of symmetry is:

- (A) $\frac{h}{4}$
 (B) $\frac{h}{3}$
 (C) $\frac{3}{4}h$
 (D) $\frac{2}{3}h$

Q9. A uniform square plate of side a has a small square of side $a/2$ removed from one of its corners. The distance of the centre of mass of the remaining sheet from the corner opposite to the cutout corner is:

- (A) $\frac{5\sqrt{2}}{12}a$
 (B) $\frac{5}{12}a$

- (C) $\frac{7\sqrt{2}}{12}a$
 (D) $\frac{\sqrt{2}}{3}a$

Q10. The distance of the centre of mass of a uniform semicircular lamina (disc) of radius R from the flat diameter is:

- (A) $\frac{2R}{\pi}$
 (B) $\frac{4R}{3\pi}$
 (C) $\frac{3R}{4\pi}$
 (D) $\frac{3R}{8}$

Q11. A thin uniform wire is bent into the shape of a semicircle of radius R lying in the xy -plane with its diameter along the x -axis and symmetric about the y -axis. The coordinates $(x_{\text{cm}}, y_{\text{cm}})$ of its centre of mass are:

- (A) $\left(0, \frac{4R}{3\pi}\right)$
 (B) $\left(\frac{2R}{\pi}, 0\right)$
 (C) $\left(0, \frac{2R}{\pi}\right)$
 (D) $(0, R)$

Q12. The centre of mass of a uniform hollow right circular cone (without base) of height h from its apex lies at a distance:

- (A) $\frac{3}{4}h$
 (B) $\frac{2}{3}h$
 (C) $\frac{1}{3}h$
 (D) $\frac{1}{2}h$

Q13. A uniform laminar sheet is in the shape of a quadrant of a circular disc of radius R lying in the first quadrant ($x \geq 0, y \geq 0$). The coordinates $(x_{\text{cm}}, y_{\text{cm}})$ are:

- (A) $\left(\frac{4R}{3\pi}, \frac{4R}{3\pi}\right)$
 (B) $\left(\frac{2R}{\pi}, \frac{2R}{\pi}\right)$
 (C) $\left(\frac{4R}{3\pi}, 0\right)$
 (D) $\left(\frac{3R}{8}, \frac{3R}{8}\right)$

Q14. A circular disc of radius R has a surface mass density $\sigma(r) = \sigma_0 \left(\frac{r}{R}\right)$, where r is the radial distance from the centre. The distance of the centre of mass of the disc from its centre is:

- (A) $\frac{2}{3}R$

- (B) $\frac{1}{2}R$
 (C) 0
 (D) $\frac{3}{4}R$

Q15. A non-uniform circular ring of radius R lies in the xy -plane centered at the origin. Its linear mass density is given by $\lambda(\theta) = \lambda_0 \cos^2 \theta$, where θ is measured from the $+x$ -axis. The location of its centre of mass is:

- (A) $\left(\frac{R}{2}, 0\right)$
 (B) $(0, 0)$
 (C) $\left(0, \frac{2R}{\pi}\right)$
 (D) $\left(\frac{2R}{\pi}, 0\right)$

Q16. A uniform solid hemisphere of radius R and a uniform solid right circular cone of base radius R and height h are joined base-to-base made of the same material. If the centre of mass of the combined body lies at the center of the common circular base, the ratio h/R must be:

- (A) $\sqrt{3}$
 (B) 2
 (C) $\sqrt{2}$
 (D) $\frac{\sqrt{3}}{2}$

Q17. Which of the following statements about the centre of mass of a continuous rigid body is INCORRECT?

- (A) The centre of mass of a body must always lie inside the physical material of the body
 (B) For a body with uniform density having a plane of symmetry, the centre of mass lies on the plane of symmetry
 (C) The position of the centre of mass is independent of the choice of coordinate axes
 (D) In a uniform gravitational field, the centre of mass coincides with the centre of gravity

Q18. A uniform cylinder of radius R and length L has a hemispherical cap of radius R of the same uniform density attached to one of its flat ends. If $L = R$, the distance of the centre of mass of the entire solid from the other flat base of the cylinder is:

- (A) $\frac{17}{20}R$
 (B) $\frac{19}{20}R$
 (C) $\frac{23}{24}R$
 (D) $\frac{11}{12}R$

Q19. A thin rod of length L has linear mass density $\lambda(x) = ax^2$ for $0 \leq x \leq L$. The position of its centre of mass from $x = 0$ is:

- (A) $\frac{2}{3}L$

(B) $\frac{4}{5}L$

(C) $\frac{1}{2}L$

(D) $\frac{3}{4}L$

Q20. A uniform rectangular sheet of dimensions $2a \times 2b$ is folded along its line of symmetry parallel to side $2b$ so

that the two halves overlap. The shift in the centre of mass of the sheet is:

(A) $\frac{a}{4}$

(B) $\frac{a}{2}$

(C) a

(D) $\frac{b}{2}$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-048)

Centre of Mass of a Rigid Body & Continuous Mass Distribution

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	A	6	A	11	C	16	A
2	C	7	B	12	B	17	A
3	D	8	C	13	A	18	A
4	A	9	A	14	C	19	D
5	D	10	B	15	B	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (A)

For a 1D continuous mass distribution:

$$x_{\text{cm}} = \frac{\int_0^L x dm}{\int_0^L dm} = \frac{\int_0^L x^2 dx}{\int_0^L x dx}$$

$$x_{\text{cm}} = \frac{L^3/3}{L^2/2} = \frac{2}{3}L$$

Sol 2. Option (C)

Element of arc $d\theta$ at angle θ has $y = R \sin \theta$:

$$y_{\text{cm}} = \frac{\int_0^\pi (R \sin \theta)(\lambda R d\theta)}{\int_0^\pi \lambda R d\theta}$$

$$y_{\text{cm}} = \frac{R[-\cos \theta]_0^\pi}{\pi} = \frac{2R}{\pi}$$

Sol 3. Option (D)

Original disc mass M at $(0, 0)$. Cutout mass $m = M/4$ with centre at $x_1 = R/2$:

$$x_{\text{cm}} = \frac{M(0) - (M/4)(R/2)}{M - M/4} = -\frac{R}{6}$$

Distance from the centre is $R/6$.

Sol 4. Option (A)

Slicing into thin discs of radius $r = \sqrt{R^2 - y^2}$:

$$y_{\text{cm}} = \frac{\int_0^R y \pi (R^2 - y^2) dy}{\int_0^R \pi (R^2 - y^2) dy} = \frac{R^4/4}{2R^3/3} = \frac{3}{8}R$$

Sol 5. Option (D)

Individual sides have COMs at $(L/2, 0)$, $(L/4, \frac{\sqrt{3}}{4}L)$, and $(3L/4, \frac{\sqrt{3}}{4}L)$.

$$y_{\text{cm}} = \frac{m(0) + 2m \left(\frac{\sqrt{3}}{4}L \right)}{3m} = \frac{\sqrt{3}}{6}L$$

Sol 6. Option (A)

For a thin hemispherical shell of radius R :

$$y_{\text{cm}} = \frac{\int_0^{\pi/2} (R \cos \theta)(2\pi R^2 \sin \theta d\theta)}{2\pi R^2}$$

$$y_{\text{cm}} = R \left[\frac{\sin^2 \theta}{2} \right]_0^{\pi/2} = \frac{R}{2}$$

Sol 7. Option (B)

Total mass:

$$M = \lambda_0 \int_0^L \left(1 + \frac{x}{L} \right) dx = \frac{3}{2} \lambda_0 L$$

First moment:

$$\int_0^L x dm = \lambda_0 \int_0^L \left(x + \frac{x^2}{L} \right) dx = \frac{5}{6} \lambda_0 L^2$$

$$x_{\text{cm}} = \frac{\frac{5}{6} \lambda_0 L^2}{\frac{3}{2} \lambda_0 L} = \frac{5}{9}L$$

Sol 8. Option (C)

Taking apex at origin $y = 0$:

$$y_{\text{cm}} = \frac{\int_0^h y^3 dy}{\int_0^h y^2 dy} = \frac{h^4/4}{h^3/3} = \frac{3}{4}h$$

Sol 9. Option (A)

Original square (M) at $(a/2, a/2)$; cutout ($M/4$) at $(3a/4, 3a/4)$:

$$x_{\text{cm}} = y_{\text{cm}} = \frac{M(a/2) - (M/4)(3a/4)}{3M/4} = \frac{5}{12}a$$

Distance from origin:

$$d = \sqrt{x_{\text{cm}}^2 + y_{\text{cm}}^2} = \frac{5\sqrt{2}}{12}a$$

Sol 10. Option (B)

For a semicircular disc:

$$y_{\text{cm}} = \frac{\int_0^R \left(\frac{2r}{\pi} \right) (\pi r dr)}{\int_0^R \pi r dr} = \frac{4R}{3\pi}$$

Sol 11. Option (C)

By symmetry $x_{\text{cm}} = 0$. For a semicircular wire of radius R , $y_{\text{cm}} = \frac{2R}{\pi}$.

Hence, coordinates are $(0, \frac{2R}{\pi})$.

Sol 12. Option (B)

Taking apex at origin $y = 0$:

$$y_{\text{cm}} = \frac{\int_0^h y^2 dy}{\int_0^h y dy} = \frac{h^3/3}{h^2/2} = \frac{2}{3}h$$

Sol 13. Option (A)

By symmetry between x and y axes for a quadrant:

$$x_{\text{cm}} = \frac{4R}{3\pi}, \quad y_{\text{cm}} = \frac{4R}{3\pi}$$

Sol 14. Option (C)

Due to complete radial symmetry of $\sigma(r)$ about the origin, the centre of mass lies at $(0, 0)$.

Sol 15. Option (B)

Integrals $\int_0^{2\pi} \cos^3 \theta d\theta = 0$ and $\int_0^{2\pi} \sin \theta \cos^2 \theta d\theta = 0$.
Thus, the centre of mass is at $(0, 0)$.

Sol 16. Option (A)

Base at $z = 0$:

Hemisphere: $V_1 = \frac{2}{3}\pi R^3$, $z_1 = -\frac{3}{8}R$.

Cone: $V_2 = \frac{1}{3}\pi R^2 h$, $z_2 = +\frac{h}{4}$.

$$V_1 z_1 + V_2 z_2 = 0 \implies \frac{h^2}{12} = \frac{R^2}{4} \implies \frac{h}{R} = \sqrt{3}$$

Sol 17. Option (A)

The centre of mass need not lie within the material of the body (e.g., ring, hollow sphere). Hence, (A) is the incorrect statement.

Sol 18. Option (A)

Base of cylinder at $z = 0$:

Cylinder: $V_c = \pi R^3$, $z_c = R/2$.

Hemisphere: $V_h = \frac{2}{3}\pi R^3$, $z_h = R + \frac{3}{8}R = \frac{11}{8}R$.

$$z_{\text{cm}} = \frac{\pi R^3 \left(\frac{R}{2}\right) + \frac{2}{3}\pi R^3 \left(\frac{11}{8}R\right)}{\pi R^3 + \frac{2}{3}\pi R^3}$$

$$z_{\text{cm}} = \frac{\frac{1}{2} + \frac{11}{12}}{1 + \frac{2}{3}} R = \frac{17/12}{5/3} R = \frac{17}{20} R$$

Sol 19. Option (D)

Given $\lambda(x) = ax^2$:

$$x_{\text{cm}} = \frac{\int_0^L x^3 dx}{\int_0^L x^2 dx} = \frac{L^4/4}{L^3/3} = \frac{3}{4}L$$

Sol 20. Option (B)

Initial COM at (a, b) . Folded half spans from $x = 0$ to $x = a$ with new COM at $(a/2, b)$.

Shift: $\Delta x = a - a/2 = a/2$.